1. Prove " Perceptron learning algorithm — proccesses of perceptron learning algorithm — step3":

$\Rightarrow$ set weighted sum function $z = w_0 \cdot x_0 + w_1 \cdot x_1 \cdots\cdots w_n x_n$

$\Rightarrow$ compute gradient descent of $z(w_0, w_1, \ldots w_n) = -\nabla z = -\begin{bmatrix} \frac{\partial z}{\partial w_0} \\ \frac{\partial z}{\partial w_1} \\ \vdots \\ \frac{\partial z}{\partial w_n} \end{bmatrix}$

$\Rightarrow$ use gradient descent method:

$$\vec{W}' = \vec{W} + \eta \cdot (-\nabla z) \longrightarrow \begin{bmatrix} w_0' \\ w_1' \\ \vdots \\ w_n' \end{bmatrix} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} + \eta \cdot \left(- \begin{bmatrix} \frac{\partial z}{\partial w_0} \\ \frac{\partial z}{\partial w_1} \\ \vdots \\ \frac{\partial z}{\partial w_n} \end{bmatrix} \right) \longrightarrow \begin{bmatrix} w_0' \\ w_1' \\ \vdots \\ w_n' \end{bmatrix} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix} + \eta \cdot \left(- \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_n \end{bmatrix} \right) \longrightarrow \left(\begin{bmatrix} w_0' = w_0' - \eta \cdot x_0 \\ w_1' = w_1' - \eta \cdot x_1 \\ \vdots \\ w_n' = w_n' - \eta \cdot x_n \end{bmatrix} \right)$$

$\Rightarrow$ $\left(\begin{bmatrix} w_0' = w_0' - \eta \cdot x_0 \\ w_1' = w_1' - \eta \cdot x_1 \\ \vdots \\ w_n' = w_n' - \eta \cdot x_n \end{bmatrix} \right)$ is the same as

step3: match predicted output: p_y & output y (ground truth):

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# if "( p_y <0 ) != ( y = +1 ) " = " ( p_y = -1 ) != ( y = +1 ) "
 ,then " add each w[i] by η * x[i]" = " w[i] += ( y * η * x[i] ), y = +1"

# if " ( p_y >0 ) != ( y = -1 )" = " ( p_y = +1 ) != ( y = -1 ) "
 ,then " subtract each w[i] by η * x[i]" = " w[i] += ( y * η * x[i] ), y = -1"
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2. Gradient Descent Method vs. Stochastic Gradient Descent Method (SGD):

A. gradient descent method would compute mean value of the gradient descent for all individual training example before updating their wei hts.

B. stochastic gradient descent method would compute value of the gradient descent for a single training example before updating its weights.